

Counterexamples to the modular restricted-invertibility conjectures

Solution packet for arXiv:2208.05223

11 August 2026

Candidate full negative answers. The commutative and noncommutative modular Bourgain–Tzafriri conjectures in arXiv:2208.05223 are false, even with determinant zero as explicitly allowed. A single fixed separable commutative C^* -algebra works in all dimensions. The construction also works over a commutative von Neumann algebra, defeating the suggested W^* -algebra fallback.

1 The conjectures

For a unital C^* -algebra $\mathcal{A}$, the source equips the left module $\mathcal{A}^d$ with

$$\langle x, y \rangle = \sum_{r=1}^d x_r y_r^*, \quad \|x\| = \|\langle x, x \rangle\|^{1/2}.$$

Its Conjecture 1.2 asserts that for every unital commutative $\mathcal{A}$ there are constants $a_0, c_0 > 0$, depending only on $\mathcal{A}$, such that every $M \in M_d(\mathcal{A})$ with unit columns and $\det M \in \mathcal{A}^\times \cup \{0\}$ has a coordinate subset σ with

$$|\sigma| \geq \frac{c_0 d}{\|M\|^2} \tag{1}$$

and

$$\left\langle \sum_{j \in \sigma} a_j M e_j, \sum_{j \in \sigma} a_j M e_j \right\rangle \geq a_0 \sum_{j \in \sigma} a_j a_j^* \quad (a_j \in \mathcal{A}). \tag{2}$$

Conjecture 1.3 makes the analogous assertion for Manin matrices over an arbitrary unital C^* -algebra, using the column determinant.

2 The simultaneous counterexample

Theorem 1. *Both Conjecture 1.2 and Conjecture 1.3 are false. More precisely, there is a fixed separable unital commutative C^* -algebra $\mathcal{A}$ such that, for every $d \geq 2$, there is $M_d \in M_d(\mathcal{A})$ satisfying*

$$\langle M_d e_j, M_d e_j \rangle = \mathbf{1}, \quad \det M_d = 0, \quad \|M_d\| = \sqrt{2},$$

but no coordinate subset of cardinality at least two satisfies (2) for any $a_0 > 0$.

The same assertion holds with $\mathcal{A} = \ell_\infty$, which is a commutative von Neumann algebra. All the matrices are Manin matrices, and their Manin column determinants are zero.

3 Idea of the proof

For $\mathcal{A} = C(K)$ a module matrix is a continuous family of ordinary scalar matrices. The conjecture demands one coordinate subset that works at every fiber. We arrange that every pair $\{j, k\}$ of

$$\|(x_j)_{j=1}^d\| := \left\| \sum_{j=1}^d x_j x_j^* \right\|^{\frac{1}{2}}, \quad \forall (x_j)_{j=1}^d \in \mathcal{A}^d.$$

In this way $\mathcal{A}^d$ becomes **standard Hilbert C^* -module** [37][50][52].

Conjecture 1.2. *[(Commutative) Modular Bourgain-Tzafriri Restricted Invertibility Conjecture] Let $\mathcal{A}$ be a unital commutative C^* -algebra and $\mathcal{I}(\mathcal{A})$ be the set of all invertible elements of $\mathcal{A}$. For $d \in \mathbb{N}$, let $\mathbb{M}_{d \times d}(\mathcal{A})$ be the set of all d by d matrices over $\mathcal{A}$. For $M \in \mathbb{M}_{d \times d}(\mathcal{A})$, let $\det(M)$ be the determinant of M . Let $\mathcal{A}^d$ be the standard (left) Hilbert C^* -module over $\mathcal{A}$ and $\{e_j\}_{j=1}^d$ be the canonical orthonormal basis for $\mathcal{A}^d$. There are universal real constants $A > 0$, $c > 0$ (A and c may depend upon C^* -algebra $\mathcal{A}$) satisfying the following property. If $d \in \mathbb{N}$ and $M \in \mathbb{M}_{d \times d}(\mathcal{A})$ with $\langle Me_j, Me_j \rangle = 1$, $\forall 1 \leq j \leq d$ and $\det(M) \in \mathcal{I}(\mathcal{A}) \cup \{0\}$, then there exists a subset $\sigma \subseteq \{1, \dots, d\}$ of cardinality*

$$\text{Card}(\sigma) \geq \frac{cd}{\|M\|^2}$$

such that

$$\sum_{j \in \sigma} \sum_{k \in \sigma} a_j \langle Me_j, Me_k \rangle a_k^* = \left\langle \sum_{j \in \sigma} a_j Me_j, \sum_{k \in \sigma} a_k Me_k \right\rangle \geq A \sum_{j \in \sigma} a_j a_j^*, \quad \forall a_j \in \mathcal{A}, \forall j \in \sigma,$$

where $\|M\|$ is the norm of the Hilbert C^ -module homomorphism defined by M as $M : \mathcal{A}^d \ni x \mapsto Mx \in \mathcal{A}^d$.*

We next formulate Conjecture [1.2] for unital C^* -algebras which need not be commutative. In the statement we use the notion of Manin matrices. We refer [17][18] for the basics of Manin matrices.

Conjecture 1.3. *[(Noncommutative) Modular Bourgain-Tzafriri Restricted Invertibility Conjecture] Let $\mathcal{A}$ be a unital C^* -algebra and $\mathcal{I}(\mathcal{A})$ be the set of all invertible elements of $\mathcal{A}$. For $d \in \mathbb{N}$, let $\mathbb{MM}_{d \times d}(\mathcal{A})$ be the set of all d by d Manin matrices over $\mathcal{A}$. For $M \in \mathbb{MM}_{d \times d}(\mathcal{A})$, let $\det^{\text{column}}(M)$ be the Manin determinant of M by column expansion. Let $\mathcal{A}^d$ be the standard (left) Hilbert C^* -module over $\mathcal{A}$ and $\{e_j\}_{j=1}^d$ be the canonical orthonormal basis for $\mathcal{A}^d$. There are universal real constants $A > 0$, $c > 0$ (A and c may depend upon C^* -algebra $\mathcal{A}$) satisfying the following property. If $d \in \mathbb{N}$ and $M \in \mathbb{MM}_{d \times d}(\mathcal{A})$ with $\langle Me_j, Me_j \rangle = 1$, $\forall 1 \leq j \leq d$ and $\det^{\text{column}}(M) \in \mathcal{I}(\mathcal{A}) \cup \{0\}$, then there exists a subset $\sigma \subseteq \{1, \dots, d\}$ of cardinality*

$$\text{Card}(\sigma) \geq \frac{cd}{\|M\|^2}$$

such that

$$\sum_{j \in \sigma} \sum_{k \in \sigma} a_j \langle Me_j, Me_k \rangle a_k^* = \left\langle \sum_{j \in \sigma} a_j Me_j, \sum_{k \in \sigma} a_k Me_k \right\rangle \geq A \sum_{j \in \sigma} a_j a_j^*, \quad \forall a_j \in \mathcal{A}, \forall j \in \sigma,$$

2

Figure 1: Conjectures 1.2 and 1.3 on page 2 of the official arXiv PDF.

columns coincides at one designated fiber. Therefore every fixed subset of two or more columns is singular somewhere.

The decisive quantitative point is that duplicating exactly one pair of otherwise orthonormal unit columns costs only a factor $\sqrt{2}$ in operator norm. Hence the conjectured size lower bound remains linear in d , while the largest uniformly invertible coordinate subset has size one.

4 Proof of Theorem 1

Let

$$\Omega = \{(d, j, k) : d \geq 2, 1 \leq j < k \leq d\}$$

with the discrete topology, let $K = \Omega \cup \{\infty\}$ be its one-point compactification, and set $\mathcal{A} = C(K)$. Thus $\mathcal{A}$ is the algebra of convergent scalar sequences; in particular it is unital, separable, and commutative.

For $1 \leq j < k \leq d$, define a scalar matrix $E_{jk}^{(d)}$ as follows. Choose a bijection

$$g : \{1, \dots, d\} \setminus \{k\} \longrightarrow \{1, \dots, d-1\}$$

and let its r th column be

$$E_{jk}^{(d)} e_r = \begin{cases} e_{g(r)}, & r \neq k, \\ e_{g(j)}, & r = k. \end{cases}$$

Here the vectors on the right are the first $d-1$ standard vectors of $\mathbb{C}^d$. Thus columns j and k coincide, while the other columns are distinct members of an orthonormal family. It follows immediately that

$$\|E_{jk}^{(d)} e_r\| = 1, \quad \det E_{jk}^{(d)} = 0, \quad \|E_{jk}^{(d)}\| = \sqrt{2}. \quad (3)$$

Indeed its nonzero singular values are $\sqrt{2}$ and 1 with multiplicity $d-2$.

For fixed d , define the $C(K)$ -valued matrix M_d fiberwise by

$$M_d(\omega) = \begin{cases} E_{jk}^{(d)}, & \omega = (d, j, k), \\ E_{12}^{(d)}, & \text{otherwise, including } \omega = \infty. \end{cases} \quad (4)$$

Only finitely many fibers differ from the value at infinity, so every entry of M_d is continuous on K . Equations (3) and pointwise evaluation give

$$\langle M_d e_r, M_d e_r \rangle = \mathbf{1}, \quad \det M_d = 0.$$

For matrices over $C(K)$ the adjointable module-operator norm is the supremum of the scalar fiber norms. Hence $\|M_d\| = \sqrt{2}$.

Now let $\sigma \subset \{1, \dots, d\}$ have at least two elements and choose $j < k$ in σ . In (2), take the constant $C(K)$ -coefficients

$$a_j = \mathbf{1}, \quad a_k = -\mathbf{1}, \quad a_r = 0 \quad (r \notin \{j, k\}).$$

At the fiber $\omega = (d, j, k)$, the left side of (2) is zero because the two selected columns coincide, whereas the right side is $2a_0 > 0$. Positivity in $C(K)$ is pointwise, so (2) fails. Consequently every admissible σ has cardinality at most one.

If Conjecture 1.2 supplied constants $a_0, c_0 > 0$ for this fixed $\mathcal{A}$, choose $d > 2/c_0$. Since $\|M_d\|^2 = 2$, equation (1) would force $|\sigma| > 1$, a contradiction. This disproves Conjecture 1.2.

Every entry of M_d lies in a commutative algebra. Therefore all Manin commutator identities hold automatically, and the Manin column determinant equals the ordinary determinant, namely zero. The same matrices consequently satisfy every hypothesis of Conjecture 1.3 and disprove it as well.

Finally, use the same bounded coordinate functions on Ω as entries in $\ell_\infty(\Omega)$. The preceding norm and point-evaluation proof is unchanged. Since $\ell_\infty(\Omega)$ is a commutative W^* -algebra, the counterexample persists in that proposed fallback class.

5 The Johnson–Lindenstrauss statement as printed

There is a separate, elementary issue in Conjecture 2.6: both its lower and upper bounds contain $1 - \varepsilon$, rather than an upper factor $1 + \varepsilon$. Taken literally, this conjecture is also false already for $\mathcal{A} = \mathbb{C}$.

Proposition 2. *Conjecture 2.6 as printed is false over the scalar algebra $\mathbb{C}$.*

Proof. Suppose its constant is $C_0 > 0$ and fix $0 < \varepsilon < 1$. In $\mathbb{C}^N$ take the $2N + 1$ points

$$0, e_1, \dots, e_N, ie_1, \dots, ie_N.$$

The two printed inequalities force equality

$$\|L(x - y)\|^2 = (1 - \varepsilon)\|x - y\|^2$$

for every pair of these points. Setting $u_j = Le_j$, the pairs $(0, e_j)$ give $\|u_j\|^2 = 1 - \varepsilon$; the pairs (e_j, e_k) and (e_j, ie_k) give respectively the vanishing of the real and imaginary parts of $\langle u_j, u_k \rangle$ for $j \neq k$. Thus $u_1, \dots, u_N$ are nonzero mutually orthogonal vectors in $\mathbb{C}^m$, so $m \geq N$.

For all sufficiently large N there is an integer

$$C_0 \varepsilon^{-2} \log(2N + 1) < m < N,$$

contradicting the asserted existence of $L \in M_{m \times N}(\mathbb{C})$. □

universal constant $C > 0$ (which may depend upon $\mathcal{A}$) satisfying the following. Let $0 < \varepsilon < 1$, $M, N \in \mathbb{N}$ and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M \in \mathcal{A}^N$. For each natural number

$$m > C \phi(\mathcal{A}, \varepsilon, M)$$

there exists a matrix $M \in \mathbb{M}_{m \times N}(\mathcal{A})$ such that

$$(1 - \varepsilon) \langle \mathbf{x}_j - \mathbf{x}_k, \mathbf{x}_j - \mathbf{x}_k \rangle \leq \langle M(\mathbf{x}_j - \mathbf{x}_k), M(\mathbf{x}_j - \mathbf{x}_k) \rangle \leq (1 + \varepsilon) \langle \mathbf{x}_j - \mathbf{x}_k, \mathbf{x}_j - \mathbf{x}_k \rangle, \quad \forall 1 \leq j, k \leq M.$$

A particular case of Problem [2.5](#) is the following conjecture.

Conjecture 2.6. (Modular Johnson-Lindenstrauss Flattening Conjecture) *Let $\mathcal{A}$ be a unital C^* -algebra. There is a universal constant $C > 0$ (which may depend upon $\mathcal{A}$) satisfying the following. Let $0 < \varepsilon < 1$, $M, N \in \mathbb{N}$ and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M \in \mathcal{A}^N$. For each natural number*

$$m > \frac{C}{\varepsilon^2} \log M,$$

there exists a matrix $M \in \mathbb{M}_{m \times N}(\mathcal{A})$ such that

$$(1 - \varepsilon) \langle \mathbf{x}_j - \mathbf{x}_k, \mathbf{x}_j - \mathbf{x}_k \rangle \leq \langle M(\mathbf{x}_j - \mathbf{x}_k), M(\mathbf{x}_j - \mathbf{x}_k) \rangle \leq (1 + \varepsilon) \langle \mathbf{x}_j - \mathbf{x}_k, \mathbf{x}_j - \mathbf{x}_k \rangle, \quad \forall 1 \leq j, k \leq M.$$

Remark 2.7. (i) *We believe that Conjecture [2.6](#) holds at least for W^* -algebras (von Neumann algebras) or C^* -algebras with IBN property.*

(ii) **Modular Welch bounds** are derived in [\[40\]](#) and various problems including modular Zauner's conjecture are stated there.

REFERENCES

- [1] Dimitris Achlioptas. Database-friendly random projections: Johnson-Lindenstrauss with binary coins. volume 66, pages 671–687. 2003. Special issue on PODS 2001 (Santa Barbara, CA).
- [9] Nir Ailon and Bernard Chazelle. Approximate nearest neighbors and the fast Johnson-Lindenstrauss transform. In

Figure 2: Conjecture 2.6 as printed on page 4; the repeated upper factor $1 + \varepsilon$ is visible.

Proposition 2 addresses only the literal statement. It makes no claim about the evidently intended variant with upper factor $1 + \varepsilon$.

6 Verification, novelty, and review

All parts of the principal construction are exact: the spectrum is fixed for every d ; only finitely many values differ from the limit fiber; the fiber norms are exactly $\sqrt{2}$; and the determinant is the zero element, not merely noninvertible. The use of constant coefficients $1, -1$ makes failure of the module order inequality visible at a single character.

The run registry, parsed corpus, exact conjecture/title searches, author/citation searches, and close searches for modular or continuous matrix-function restricted invertibility were checked through 11 August 2026. No later direct solution or counterexample was found. The closest located work, Fan et al. [2], allows the selected subspace to vary with the fiber; it does not produce the one fixed coordinate subset demanded by the source conjectures. Novelty confidence is moderate.

Human review should prioritize the quantifier that a_0, c_0 may depend on $\mathcal{A}$ but not on d ; this is why one fixed infinite-dimensional algebra is essential. The restricted-invertibility counterexamples are independent of the likely typographical error in Conjecture 2.6 and give complete negative answers to Conjectures 1.2 and 1.3 as written.

References

- [1] K. M. Krishna, *Modular Bourgain–Tzafriri Restricted Invertibility Conjectures and Johnson–Lindenstrauss Flattening Conjecture*, arXiv:2208.05223 (2022).
- [2] A. Fan, J. Montemurro, P. Motakis, N. Praveen, A. Rusonik, P. Skoufranis, and N. Tobin, *Restricted invertibility of continuous matrix functions*, *Operators and Matrices* **17** (2023), 1113–1147; arXiv:2201.04238.